🎓 Student Performance Analysis Using Excel





📌 Project Overview

This project analyzes students' academic performance using Microsoft Excel. The goal is to evaluate student marks, identify performance trends across subjects, and generate insights that help educators understand academic outcomes.

It demonstrates beginner-level data analysis skills: data organization, Excel formulas, data visualization, and reporting.

🎯 Objectives

- Analyze students' marks across different subjects
- Calculate average marks for individual students and subjects
- Identify the highest and lowest scores
- Compare academic performance across subjects
- Calculate the pass percentage
- Create charts to visualize student performance
- Present meaningful insights through an Excel dashboard

🛠️ Tools and Technologies

- Microsoft Excel
- Excel Formulas
- Sorting and Filtering
- Conditional Formatting
- PivotTables
- Charts and Data Visualization

📂 Dataset Description

The dataset contains student information and the marks obtained in different subjects.

Student ID	Student Name	Mathematics	Science	English	Total Marks	Average Marks	Result
101	Aarav	85	90	78	253	84.33	Pass
102	Diya	92	88	95	275	91.67	Pass
103	Rahul	45	55	60	160	53.33	Pass
104	Priya	35	40	38	113	37.67	Fail
105	Arjun	76	82	80	238	79.33	Pass
106	Ananya	88	91	85	264	88.00	Pass
107	Rohan	55	48	52	155	51.67	Pass
108	Sneha	95	96	92	283	94.33	Pass
109	Karan	30	35	42	107	35.67	Fail
110	Meera	80	75	89	244	81.33	Pass

> **Note:** This is sample data created for practice purposes. It does not represent real students.

🧹 Data Preparation

The following steps are performed in Excel:

1. Check for missing values
2. Identify and remove duplicate records, if any
3. Ensure marks are stored as numeric values
4. Verify that all marks fall within the valid range of 0–100
5. Calculate total and average marks
6. Assign Pass or Fail status using a defined passing criterion

Passing criterion: A student must score at least **40 marks in every subject** to pass.

📊 Excel Formulas Used

Assuming Mathematics, Science, and English marks are in columns C, D, and E, with data in rows 2–11:

Purpose	Formula
Total marks	`=SUM(C2:E2)`
Average marks	`=AVERAGE(C2:E2)`
Highest score	`=MAX(C2:E11)`
Lowest score	`=MIN(C2:E11)`
Pass / Fail	`=IF(AND(C2>=40,D2>=40,E2>=40),"Pass","Fail")`
Overall pass percentage	`=COUNTIF(H2:H11,"Pass")/COUNTA(H2:H11)*100`

> For the pass percentage, the Result column is column H. If you remove `*100` from the formula, format the cell as a percentage instead.

📈 Analysis Performed

- Student-wise total and average marks
- Subject-wise average scores
- Highest and lowest scores by subject
- Comparison of student performance across subjects
- Number of students who passed and failed
- Overall pass percentage
- Identification of high-performing students

📉 Dashboard Features

The interactive Excel dashboard includes:

- **Total Students:** Number of students included in the analysis
- **Average Marks:** Average score across all student-subject marks
- **Highest Marks:** Highest individual subject score

- **Pass Percentage:** Percentage of students who passed all three subjects
- **Subject Performance Chart:** Comparison of average marks across Mathematics, Science, and English
- **Student Performance Chart:** Comparison of average marks by student
- **Pass/Fail Chart:** Distribution of students by result

🔍 Key Findings

Based on the sample dataset:

Metric	Value
Total students	10
Overall average across all subject marks	69.73
Highest individual subject score	96
Lowest individual subject score	30
Students passed	8
Students failed	2
Overall pass percentage	80%

Subject-wise averages:

Subject	Average
Mathematics	68.1
Science	70.0
English	71.1

Mathematics has the lowest average, while English has the highest.

💡 Recommendations

- Provide additional academic support to students who are struggling

- Identify subjects where students have lower average scores
- Encourage high-performing students to maintain their progress
- Monitor student performance regularly to identify changes over time
- Use performance reports to support teaching and learning decisions

These recommendations should be adapted to the actual findings from the completed analysis.

📁 Project Structure

```
Student-Performance-Analysis/
│
├── README.md
├── Student_Performance_Analysis.xlsx
└── dashboard_screenshot.png
```

- `README.md` — Project documentation
- `Student_Performance_Analysis.xlsx` — Workbook containing the dataset, calculations, analysis, and dashboard
- `dashboard_screenshot.png` — Screenshot of the completed dashboard

🚀 How to Use This Project

1. Download or clone the repository
2. Open `Student_Performance_Analysis.xlsx` in Microsoft Excel
3. Review the student dataset
4. Explore the formulas and calculated columns
5. Examine the charts and dashboard
6. Modify the sample data to practice additional analyses

📚 Skills Demonstrated

- Data cleaning and preparation
- Excel formulas and functions
- Descriptive statistics
- Data analysis

- Data visualization
- Dashboard development
- Business-style reporting and communication of findings

🏁 Conclusion

This project shows how Microsoft Excel can be used to organize, analyze, and visualize academic data. It gives beginner-level experience in calculating performance metrics, comparing results, and communicating findings through charts and dashboards, and it is a good foundation for learning tools like SQL, Power BI, and Python.

Project Category: Data Analytics
Difficulty Level: Beginner
Tool Used: Microsoft Excel
Author: Your Name